# TEST REPORT

| | |
|---|---|
| **Application No.:** | KSEM2204000431AT |
| **Applicant:** | Richbeam (Suzhou) Technology Co.,LTD |
| **Address of Applicant:** | NO.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China |
| **Manufacturer:** | Richbeam (Suzhou) Technology Co.,LTD |
| **Address of Manufacturer:** | NO.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China |
| **Factory:** | Richbeam (Suzhou) Technology Co.,LTD |
| **Address of Factory:** | NO.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China |
| **Equipment Under Test (EUT):** | |
| **EUT Name:** | 2D TOF LIDAR |
| **Model No.:** | LakiBeam 1, LakiBeam 1L ♣ |
| ♣ | Please refer to section 2 of this report which indicates which model was actually tested and which were electrically identical. |
| **Trade Mark:** | RICHBEAM |
| **Standard(s) :** | EN IEC 61326-1: 2021 |
| **Date of Receipt:** | 2022-04-12 |
| **Date of Test:** | 2022-04-25 to 2022-04-28 |
| **Date of Issue:** | 2022-05-05 |
| **Test Result:** | **Pass*** |

* In the configuration tested, the EUT complied with the standards specified above.

Eric Lin
EMC Laboratory Manager

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| Revision Record | | | | |
|---|---|---|---|---|
| **Version** | **Chapter** | **Date** | **Modifier** | **Remark** |
| 01 | | 2022-05-05 | | Original |
| | | | | |
| | | | | |

| | | | | |
|---|---|---|---|---|
| **Authorized for issue by:** | | | | |
| | | Lee Li<br>**Lee Li/Project Engineer** | | |
| | | Eric Lin<br>**Eric Lin/Reviewer** | | |

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# 2 Test Summary

| Emission Part | | | | |
|---|---|---|---|---|
| **Item** | **Standard** | **Method** | **Requirement** | **Result** |
| Conducted Emissions at AC Mains Power Port (150kHz-30MHz) | EN IEC 61326-1: 2021 | EN 55011:2016+A11:2020 | Group 2 Class B | Pass |
| Radiated Emission(30MHz-1GHz) | | EN 55011:2016+A11:2020 | Group 2 Class B | Pass |
| Radiated Emissions (Magnetic field Strength)(150kHz-30MHz) | | EN 55011:2016+A11:2020 | Group 2 Class B | Pass |
| Harmonic Current Emission | | EN IEC 61000-3-2:2019+A1:2021 | Class A | Pass |
| Voltage Fluctuations and Flicker | | EN 61000-3-3:2013+A1:2019 | Clause 5 | Pass |

| Immunity Part | | | | |
|---|---|---|---|---|
| **Item** | **Standard** | **Method** | **Requirement** | **Result** |
| Conducted Immunity at AC Mains Power Port (150kHz-80MHz) | EN IEC 61326-1: 2021 | EN 61000-4-6:2014 | 3 Vrms: 0.15MHz - 80MHz,3Vrms,80%,1kHz Amp. Mod. | Pass |
| Conducted Immunity at Signal Port (150kHz-80MHz) | | EN 61000-4-6:2014 | 3 Vrms: 0.15MHz - 80MHz,3 Vrms,80%,1kHz Amp. Mod. | Pass |
| Electrical Fast Transients Burst at AC Mains Power Port | | EN 61000-4-4:2012 | 1kV,5/50ns Tr/Td,5kHz Repetition Frequency | Pass |
| Electrical Fast Transients&Burst at Signal Port | | EN 61000-4-4:2012 | 1kV,5/50ns Tr/Td,5kHz Repetition Frequency | Pass |
| Electrostatic Discharge | | EN 61000-4-2:2009 | 4kV Contact Discharge, 8kV Air Discharge | Pass |
| Radiated Immunity(80MHz-6GHz) | | EN IEC 61000-4-3: 2020 | 3V/m 80MHz to 1GHz, 3V/m 1.4GHz to 6GHz, Modulation: 1kHz,80% Amp. Mod, 1% increment | Pass |
| Surge at AC Mains Power Port | | EN 61000-4-5:2014 +A1:2017 | 1.2/50μs Tr/Td, 0.5kV Line to Line, 1.0kV Line to Ground | Pass |
| Surge at Signal Port | | EN 61000-4-5:2014 +A1:2017 | 1.2/50μs Tr/Td, 0.5kV Line to Line, 1.0kV Line to Ground | Pass |

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| Immunity Part | | | | |
|---|---|---|---|---|
| **Item** | **Standard** | **Method** | **Requirement** | **Result** |
| Voltage Dips and Interruptions | | EN IEC 61000-4-11:2020 | 0% of UT (Supply Voltage) for 0.5 Cycle:B; 0% of UT for 1 Cycle:B; 0% of UT for 250 Cycles:C; 70% of UT for 25 Cycles:C; | Pass |
| Power Frequency Magnetic Field | | EN 61000-4-8:2010 | 50Hz, 60Hz 3A/m | Pass |

**Declaration of EUT Family Grouping:**

There are series models mentioned in this report, and they are the Identical in electrical and electronic characters.Only the model LakiBeam 1L was tested since their differences were the model number, trade name,Color and appearance.

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# 3 Contents

Page

1 COVER PAGE ..... 1

2 TEST SUMMARY ..... 3

3 CONTENTS ..... 5

4 GENERAL INFORMATION ..... 8

4.1 DETAILS OF E.U.T. ..... 8
4.2 DESCRIPTION OF SUPPORT UNITS ..... 8
4.3 MEASUREMENT UNCERTAINTY ..... 8
4.4 TEST LOCATION ..... 9
4.5 TEST FACILITY ..... 9
4.6 DEVIATION FROM STANDARDS ..... 9
4.7 ABNORMALITIES FROM STANDARD CONDITIONS ..... 9

5 EQUIPMENT LIST ..... 10

6 EMISSION TEST RESULTS ..... 16

6.1 CONDUCTED EMISSIONS AT AC MAINS POWER PORT (150KHZ-30MHZ) ..... 16
*6.1.1 E.U.T. Operation ..... 16*
*6.1.2 Test Mode Description ..... 16*
*6.1.3 Test Setup Diagram ..... 16*
*6.1.4 Measurement Procedure and Data ..... 16*
6.2 RADIATED EMISSION(30MHZ-1GHZ) ..... 19
*6.2.1 E.U.T. Operation ..... 19*
*6.2.2 Test Mode Description ..... 19*
*6.2.3 Test Setup Diagram ..... 19*
*6.2.4 Measurement Procedure and Data ..... 19*
6.3 RADIATED EMISSIONS (MAGNETIC FIELD STRENGTH)(150KHZ-30MHZ) ..... 22
*6.3.1 E.U.T. Operation ..... 22*
*6.3.2 Test Mode Description ..... 22*
*6.3.3 Test Setup Diagram ..... 22*
*6.3.4 Measurement Procedure and Data ..... 22*
6.4 HARMONIC CURRENT EMISSION ..... 25
*6.4.1 E.U.T. Operation ..... 25*
*6.4.2 Test Mode Description ..... 25*
*6.4.3 Test Setup Diagram ..... 25*
*6.4.4 Measurement Procedure and Data ..... 25*
6.5 VOLTAGE FLUCTUATIONS AND FLICKER ..... 28
*6.5.1 E.U.T. Operation ..... 28*
*6.5.2 Test Mode Description ..... 28*
*6.5.3 Test Setup Diagram ..... 28*
*6.5.4 Measurement Procedure and Data ..... 28*

7 IMMUNITY TEST RESULTS ..... 29

7.1 CONDUCTED IMMUNITY AT AC MAINS POWER PORT (150KHZ-80MHZ) ..... 30

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*7.1.1 Test Setup Diagram ..... 30*
*7.1.2 E.U.T. Operation ..... 30*
*7.1.3 Test Mode Description ..... 30*
*7.1.4 Test Condition and Results: ..... 30*
7.2 CONDUCTED IMMUNITY AT SIGNAL PORT (150KHZ-80MHZ) ..... 31
*7.2.1 Test Setup Diagram ..... 31*
*7.2.2 E.U.T. Operation ..... 31*
*7.2.3 Test Mode Description ..... 31*
*7.2.4 Test Condition and Results: ..... 31*
7.3 ELECTRICAL FAST TRANSIENTS BURST AT AC MAINS POWER PORT ..... 32
*7.3.1 Test Setup Diagram ..... 32*
*7.3.2 E.U.T. Operation ..... 32*
*7.3.3 Test Mode Description ..... 32*
*7.3.4 Test Condition and Results: ..... 32*
7.4 ELECTRICAL FAST TRANSIENTS&BURST AT SIGNAL PORT ..... 33
*7.4.1 Test Setup Diagram ..... 33*
*7.4.2 E.U.T. Operation ..... 33*
*7.4.3 Test Mode Description ..... 33*
*7.4.4 Test Condition and Results: ..... 33*
7.5 ELECTROSTATIC DISCHARGE ..... 34
*7.5.1 Test Setup Diagram ..... 34*
*7.5.2 E.U.T. Operation ..... 34*
*7.5.3 Test Mode Description ..... 34*
*7.5.4 Test Condition and Results: ..... 34*
7.6 RADIATED IMMUNITY(80MHZ-6GHZ) ..... 36
*7.6.1 Test Setup Diagram ..... 36*
*7.6.2 E.U.T. Operation ..... 36*
*7.6.3 Test Mode Description ..... 36*
*7.6.4 Test Condition and Results: ..... 36*
7.7 SURGE AT AC MAINS POWER PORT ..... 38
*7.7.1 Test Setup Diagram ..... 38*
*7.7.2 E.U.T. Operation ..... 38*
*7.7.3 Test Mode Description ..... 38*
*7.7.4 Test Condition and Results: ..... 38*
7.8 SURGE AT SIGNAL PORT ..... 40
*7.8.1 Test Setup Diagram ..... 40*
*7.8.2 E.U.T. Operation ..... 40*
*7.8.3 Test Mode Description ..... 40*
*7.8.4 Test Condition and Results: ..... 40*
7.9 VOLTAGE DIPS AND INTERRUPTIONS ..... 41
*7.9.1 Test Setup Diagram ..... 41*
*7.9.2 E.U.T. Operation ..... 41*
*7.9.3 Test Mode Description ..... 41*
*7.9.4 Test Condition and Results: ..... 41*
7.10 POWER FREQUENCY MAGNETIC FIELD ..... 43
*7.10.1 Test Setup Diagram ..... 43*
*7.10.2 E.U.T. Operation ..... 43*
*7.10.3 Test Mode Description ..... 43*
*7.10.4 Test Condition and Results: ..... 43*
**8 TEST SETUP PHOTO ..... 44**

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9 EUT CONSTRUCTIONAL DETAILS (EUT PHOTOS) ..... 44

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# 4 General Information

## 4.1 Details of E.U.T.

| | |
|---|---|
| Power supply: | DC 9~36V |
| Test Voltage: | AC 230V/50Hz |

## 4.2 Description of Support Units

| Description | Manufacturer | Model No. | Serial No. |
|---|---|---|---|
| AC Adapter | DVE | DSA-12G-12FEU | / |
| Notebook | XIAOMI | Pro15 | 00342-35744-35614-AAOEM |

## 4.3 Measurement Uncertainty

| No. | Item | Measurement Uncertainty |
|---|---|---|
| 1 | Conducted Emission at mains port using AMN | 2.4dB (9kHz to 150kHz) |
| | | 2.2dB (150kHz to 30MHz) |
| 2 | Conducted Emission at telecommunication port using AAN | 4.0 dB (150kHz to 30MHz) |
| 3 | Radiated Power | 3.2dB |
| 4 | Radiated Emission (10m) | 4.1 dB |
| 5 | Radiated Emission (3m) | 4.6 dB (30MHz-1GHz) |
| | | 5.0dB (1GHz-6GHz) |
| | | 5.2dB (6GHz-18GHz) |
| | | 5.3dB (18GHz-40GHz) |

Note: The measurement uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.

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## 4.4 Test Location

All tests were performed at:

Compliance Certification Services (Kunshan) Inc.

No.10 Weiye Rd, Innovation park, Eco&Tec, Development Zone, Kunshan City, Jiangsu, China.

Tel: +86 512 5735 5888 Fax: +86 512 5737 0818

No tests were sub-contracted.

## 4.5 Test Facility

The test facility is recognized, certified, or accredited by the following organizations:

**• CNAS (No. CNAS L4354)**

CNAS has accredited Compliance Certification Services (Kunshan) Inc. to ISO/IEC 17025:2017 General Requirements for the Competence of Testing and Calibration Laboratories (CNAS-CL01 Accreditation Criteria for the Competence of Testing and Calibration Laboratories) for the competence in the field of testing.

**• A2LA (Certificate No. 2541.01)**

Compliance Certification Services (Kunshan) Inc. is accredited by the American Association for Laboratory Accreditation (A2LA). Certificate No. 2541.01.

**• FCC (Designation Number: CN1172)**

Compliance Certification Services (Kunshan) Inc. has been recognized as an accredited testing laboratory.

Designation Number: CN1172.

**• ISED (CAB identifier: CN0072)**

Compliance Certification Services (Kunshan) Inc. has been recognized by Innovation, Science and Economic Development Canada (ISED) as an accredited testing laboratory.

Company Number: 2324E

**• VCCI (Member No.: 1938)**

The 3m and 10m Semi-anechoic chamber and Shielded Room of Compliance Certification Services (Kunshan) Inc. has been registered in accordance with the Regulations for Voluntary Control Measures with Registration No.: R-20134, R-11600, C-11707, T-11499, G-10216 respectively.

## 4.6 Deviation from Standards

None

## 4.7 Abnormalities from Standard Conditions

None

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# 5 Equipment List

| Conducted Emissions at AC Mains Power Port (150kHz-30MHz) | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| EMI TEST RECEIVER | R&S | ESCI | 100781 | 01/22/2022 | 01/21/2023 |
| TWO-LINE V-NETWORK | R&S | ENV216 | 101604 | 10/12/2021 | 10/11/2022 |
| V (V-LISN) | SCHWARZBECK | NNLK 8129 | 8129-143 | 10/12/2021 | 10/11/2022 |
| Pulse LIMITER | R&S | ESH3-Z2 | 100609 | 01/22/2022 | 01/21/2023 |
| Software | Faratronic | EZ_EMC-v 3A1 | N/A | N/A | N/A |

| Radiated Emission(30MHz-1GHz) | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| EMI Test Receiver | R&S | ESCI | 101378 | 10/11/2021 | 10/10/2022 |
| Antenna | TESEQ | CBL 6112D | 35403 | 03/05/2022 | 03/04/2024 |
| Spectrum Analyzer | R&S | FSU26 | 200789 | 04/01/2022 | 03/31/2023 |
| Preamplifier | PANSHAN TECHNOLOGY | LNA:1~18G | / | 01/22/2022 | 01/21/2023 |
| Horn-antenna | SCHWARZBECK | BBHA9120D | 266 | 03/02/2022 | 03/01/2023 |
| Signal Analyzer | R&S | FSV40 | 101493 | 10/11/2021 | 10/10/2022 |
| Amplifier | COM-POWER | PAM-840A | 461332 | 01/22/2022 | 01/21/2023 |
| Broad-Band Horn Antenna | SCHWARZBECK | BBHA 9170 | 9170-171 | 03/22/2022 | 03/21/2023 |
| Software | Faratronic | EZ_EMC v 3A1 | N/A | N/A | N/A |

| Radiated Emissions (Magnetic field Strength)(150kHz-30MHz) | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| EMI TEST RECEIVER | R&S | ESCI | KS301101 | 01/22/2022 | 01/21/2023 |
| Pulse LIMITER | R&S | ESH3-Z2 | ESH3-Z2 | KUS1902E001 | 01/22/2022 |

| Harmonic Current Emission | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| AC Power Source | SCHAFFNER | NSG 1007 | 54788 | 10/18/2021 | 10/17/2022 |
| Software | TESEQ | CTS4-v 4.24.0 | N/A | N/A | N/A |
| Harmonic/Flicker Analyzer | KIKUSUI | KHA3000 | AS000065 | 04/01/2022 | 03/31/2023 |
| Line Impedance Network | KIKUSUI | SPEC71116 | / | 04/01/2022 | 03/31/2023 |
| Switcher | KIKUSUI | SPEC71092 | / | 08/24/2021 | 08/23/2022 |

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| | | | | | |
|---|---|---|---|---|---|
| AC Power Supply(Master) | KIKUSUI | PCR24000WE 2 | AR001701 | 04/01/2022 | 03/31/2023 |
| AC Power Supply(Slave) | KIKUSUI | PCR24000WE 2 | AN002885 | 04/01/2022 | 03/31/2023 |
| Software | KIKUSUI | HarmoCapture 3-vv 2.5.2.00 | N/A | N/A | N/A |

**Voltage Fluctuations and Flicker**

| Equipment | Manufacturer | Model No | Inventory No | Cal Date | Cal Due Date |
|---|---|---|---|---|---|
| Harmonic & Flicker Tester | SCHAFFNER | CCN 1000-1 | 72585 | 10/18/2021 | 10/17/2022 |
| AC Power Source | SCHAFFNER | NSG 1007 | 54788 | 10/18/2021 | 10/17/2022 |
| Software | TESEQ | CTS4-v 4.24.0 | N/A | N/A | N/A |
| Harmonic/Flicker Analyzer | KIKUSUI | KHA3000 | AS000065 | 04/01/2022 | 03/31/2023 |
| Line Impedance Network | KIKUSUI | SPEC71116 | / | 04/01/2022 | 03/31/2023 |
| Switcher | KIKUSUI | SPEC71092 | / | 08/24/2021 | 08/23/2022 |
| AC Power Supply(Master) | KIKUSUI | PCR24000WE 2 | AR001701 | 04/01/2022 | 03/31/2023 |
| AC Power Supply(Slave) | KIKUSUI | PCR24000WE 2 | AN002885 | 04/01/2022 | 03/31/2023 |
| Software | KIKUSUI | HarmoCapture 3-vv 2.5.2.00 | N/A | N/A | N/A |

**Conducted Immunity at AC Power Port (150kHz-80MHz)**

| Equipment | Manufacturer | Model No | Inventory No | Cal Date | Cal Due Date |
|---|---|---|---|---|---|
| Test System for Conducted and Radiated Immunity | TESEQ | NSG 4070B | 44279 | 01/22/2022 | 01/21/2023 |
| Amplifier | TESEQ | SCCXE75 | U2060-0916 | 01/22/2022 | 01/21/2023 |
| EM-Koppelzange | SCHAFFNER | KEMZ 801 | 17629 | 01/22/2022 | 01/21/2023 |
| Attenuator | EURO MC | 7860 ORGEVAL | 2226 | 04/01/2022 | 03/31/2023 |
| Directional Coupler | HIGH POWER | C21A8 | 450 | 10/12/2021 | 10/11/2022 |
| CDN（Coupling and Decoupling Network) | SCHAFFNER | CDN M216 | 16399 | 04/01/2022 | 03/31/2023 |
| CDN（Coupling and Decoupling Network) | SCHAFFNER | CDN M316 | 16935 | 04/01/2022 | 03/31/2023 |
| CDN（Coupling and Decoupling Network) | TESEQ | CDN M116 | 35358 | 04/01/2022 | 03/31/2023 |
| CDN | TESEQ | CDN M432 | 55985 | 01/22/2022 | 01/21/2023 |
| CDN | TESEQ | CDN M432-3LN | 56245 | 01/22/2022 | 01/21/2023 |

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| | | | | | |
|---|---|---|---|---|---|
| CDN | TESEQ | CDN M532 | 56180 | 01/26/2022 | 01/25/2023 |
| CDN | TESEQ | CDN M232 | 45974 | 04/01/2022 | 03/31/2023 |
| CDN | TESEQ | CDN M332 | 45360 | 04/01/2022 | 03/31/2023 |
| Software | TESEQ | NSG 4070-v 1.3.0.1 | N/A | N/A | N/A |

| **Conducted Immunity at Signal Port (150kHz-80MHz)** | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| Test System for Conducted and Radiated Immunity | TESEQ | NSG 4070B | 44279 | 01/22/2022 | 01/21/2023 |
| Amplifier | TESEQ | SCCXE75 | U2060-0916 | 01/22/2022 | 01/21/2023 |
| EM-Koppelzange | SCHAFFNER | KEMZ 801 | 17629 | 01/22/2022 | 01/21/2023 |
| Attenuator | EURO MC | 7860 ORGEVAL | 2226 | 04/01/2022 | 03/31/2023 |
| Directional Coupler | HIGH POWER | C21A8 | 450 | 10/12/2021 | 10/11/2022 |
| CDN（Coupling and Decoupling Network) | SCHAFFNER | CDN M216 | 16399 | 04/01/2022 | 03/31/2023 |
| CDN（Coupling and Decoupling Network) | SCHAFFNER | CDN M316 | 16935 | 04/01/2022 | 03/31/2023 |
| CDN（Coupling and Decoupling Network) | TESEQ | CDN S751 | 31084 | 04/01/2022 | 03/31/2023 |
| CDN（Coupling and Decoupling Network) | TESEQ | CDN M116 | 35358 | 04/01/2022 | 03/31/2023 |
| CDN | TESEQ | CDN T2-10S | 39812 | 04/01/2022 | 03/31/2023 |
| CDN | TESEQ | CDN T4-10S | 37195 | 04/01/2022 | 03/31/2023 |
| CDN | TESEQ | CDN T8-10S | 42002 | 04/01/2022 | 03/31/2023 |
| Current Clamp | TESEQ | IP-DR250 | 1609-1936 | 04/01/2022 | 03/31/2023 |
| CDN | TESEQ | CDN M432 | 55985 | 01/22/2022 | 01/21/2023 |
| CDN | TESEQ | CDN M432-3LN | 56245 | 01/22/2022 | 01/21/2023 |
| CDN | TESEQ | CDN M532 | 56180 | 01/26/2022 | 01/25/2023 |
| CDN | TESEQ | CDN M232 | 45974 | 04/01/2022 | 03/31/2023 |
| CDN | TESEQ | CDN M332 | 45360 | 04/01/2022 | 03/31/2023 |
| Software | TESEQ | NSG 4070-v 1.3.0.1 | N/A | N/A | N/A |

| **Electrical Fast Transients Burst at AC Mains Power Port** | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| EMC Immunity Tester | EMC PARTNER | Transient2000 | TRA2006_F-S-T-D-R-1503 | 12/19/2021 | 12/18/2022 |
| Coupling Network | EMC PARTNER | CN-EFT1000 | CN- | 12/19/2021 | 12/18/2022 |

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| | | | | | |
|---|---|---|---|---|---|
| | | | EFT1000_1535 | | |
| Burst Generator | SANKI | EFT-0404S | 040720051E | 04/01/2022 | 03/31/2023 |
| Coupling and Decoupling Network | SANKI | CDN-4350 | 043520001J | 04/01/2022 | 03/31/2023 |

**Electrical Fast Transients&Burst at Signal Port**

| Equipment | Manufacturer | Model No | Inventory No | Cal Date | Cal Due Date |
|---|---|---|---|---|---|
| EMC Immunity Tester | EMC PARTNER | Transient2000 | TRA2006_F-S-T-D-R-1503 | 12/19/2021 | 12/18/2022 |
| Coupling Network | EMC PARTNER | CN-EFT1000 | CN-EFT1000_1535 | 12/19/2021 | 12/18/2022 |
| Burst Generator | SANKI | EFT-0404S | 040720051E | 04/01/2022 | 03/31/2023 |
| Coupling and Decoupling Network | SANKI | CDN-4350 | 043520001J | 04/01/2022 | 03/31/2023 |

**Electrostatic Discharge**

| Equipment | Manufacturer | Model No | Inventory No | Cal Date | Cal Due Date |
|---|---|---|---|---|---|
| ESD Simulator | EM TEST | DITO 509030 | V0936105118 | 01/26/2022 | 01/25/2023 |
| ESD Simulator | THERMO | MINIZAP | 1405181 | 10/14/2021 | 10/13/2022 |
| ESD Generator | TESEQ AG | NSG 435 | 6172 | 02/03/2021 | 02/02/2022 |
| ESD Simulator | NoiseKen | ESS-2002 | ESS06Y6297 | 10/12/2021 | 10/11/2022 |

**Radiated Immunity(80MHz-6GHz)**

| Equipment | Manufacturer | Model No | Inventory No | Cal Date | Cal Due Date |
|---|---|---|---|---|---|
| Synthesized Signal Generator | AGILENT | 83732B | US37101915 | 01/22/2022 | 01/21/2023 |
| Laser probe interface | AR Worldwide | FI7000 | 354057 | 04/01/2022 | 03/31/2023 |
| E-Field Sensor | AR Worldwide | FL7006 100K-6G | 342390 | 04/01/2022 | 03/31/2023 |
| Amplifier Research (80~1000MHz 150w) | AR Worldwide | 150W1000M1 | 320947 | 10/18/2021 | 10/17/2022 |
| Amplifier Research (1~6GHz 50w) | AR Worldwide | 50S1G6M1 | 342706 | 10/18/2021 | 10/17/2022 |
| Dual Directional Coupler (1-11G) | AR Worldwide | C1-A47NFNF 35dB | 1 | N.C.R | N.C.R |
| Dual Directional Coupler (80~1000MHz 400w) | AR Worldwide | DC6180 | 302211 | N.C.R | N.C.R |
| RF POWER METER | BOONTON | 4232A-01 | 1614 | 01/22/2022 | 01/21/2023 |
| POWER SENSOR | BOONTON | 51085 | 33488 | 01/22/2022 | 01/21/2023 |

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| | | | | | |
|---|---|---|---|---|---|
| POWER SENSOR | BOONTON | 51085 | 33489 | 01/22/2022 | 01/21/2023 |
| Antenna | AR Worldwide | TP1000A | 302624 | N.C.R | N.C.R |
| Software | AR | emc ware-v 3.2.0.4 | N/A | N/A | N/A |

| **Surge at AC Mains Power Port** | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| EMC Immunity Tester | EMC PARTNER | TRA2006 | F-S-T-D-R-1503 | 12/19/2021 | 12/18/2022 |
| Coupling and Decoupling Network | EMC PARTNER | CDN-UTP8 | 1511 | 12/19/2021 | 12/18/2022 |
| Surge Generator | SANKI | LSG-0506S | 050920052J | 04/01/2022 | 03/31/2023 |
| Coupling and Decoupling Network | SANKI | CDN-5350 | 053520001J | 06/08/2021 | 06/07/2022 |

| **Surge at Signal Port** | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| EMC Immunity Tester | EMC PARTNER | TRA2006 | F-S-T-D-R-1503 | 12/19/2021 | 12/18/2022 |
| Coupling and Decoupling Network | EMC PARTNER | CDN-UTP8 | 1511 | 12/19/2021 | 12/18/2022 |
| Surge Generator | SANKI | LSG-0506S | 050920052J | 04/01/2022 | 03/31/2023 |
| Coupling and Decoupling Network | SANKI | CDN-5350 | 053520001J | 06/08/2021 | 06/07/2022 |

| **Voltage Dips and Interruptions** | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| EMC Immunity Tester | EMC PARTNER | TRA2006 | TRA2006_F-S-T-D-R-1503 | 12/19/2021 | 12/18/2022 |
| Switcher | KIKUSUI | SPEC71092 | / | 08/24/2021 | 08/23/2022 |
| AC Power Supply(Master) | KIKUSUI | PCR24000WE2 | AR001701 | 04/01/2022 | 03/31/2023 |
| AC Power Supply(Slave) | KIKUSUI | PCR24000WE2 | AN002885 | 04/01/2022 | 03/31/2023 |
| Software | KIKUSUI | Quick Immunity Sequencer 2-v 4.0.3.02 | N/A | N/A | N/A |

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| Power Frequency Magnetic Field | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| EMC Immunity Tester | EMC PARTNER | TRA2006 | TRA2006_F-S-T-D-R-1503 | 12/19/2021 | 12/18/2022 |
| Inductive Standard Coil | EMC PARTNER | MF1000-1 | MF1000-1-1514 | 12/19/2021 | 12/18/2022 |

| General used equipment | | | | | |
|---|---|---|---|---|---|
| **Equipment** | **Manufacturer** | **Model No** | **Inventory No** | **Cal Date** | **Cal Due Date** |
| Digital Pressure Meter | Mengde | DYM3 | CZ750023 | 02/01/2022 | 01/31/2023 |
| Temperature & Humidity Recorder | Anymetre | TH603 | CZ720001-1<br>CZ720001-2<br>CZ720001-3<br>CZ720001-4<br>CZ720001-5<br>CZ720001-6<br>CZ720001-7 | 10/14/2021 | 10/13/2022 |

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# 6 Emission Test Results

## 6.1 Conducted Emissions at AC Mains Power Port (150kHz-30MHz)

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 55011:2016+A11:2020

### 6.1.1 E.U.T. Operation

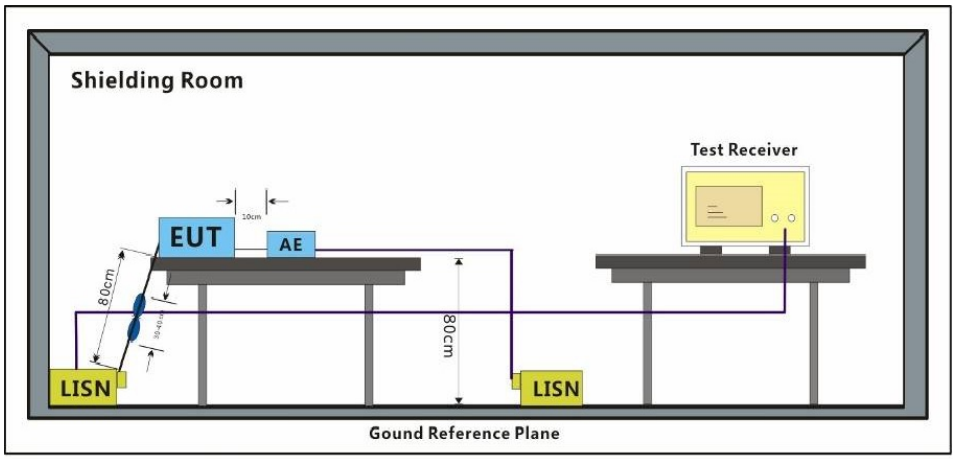
Operating Environment:

Temperature: 26.6 °C Humidity: 55.9 % RH Atmospheric Pressure: 1010 mbar

### 6.1.2 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 6.1.3 Test Setup Diagram




### 6.1.4 Measurement Procedure and Data

Frequency range: 150kHz-30MHz

An initial pre-scan was performed with peak detector. Quasi-Peak or Average measurement were performed at the frequencies with maximized peak emission were detected.

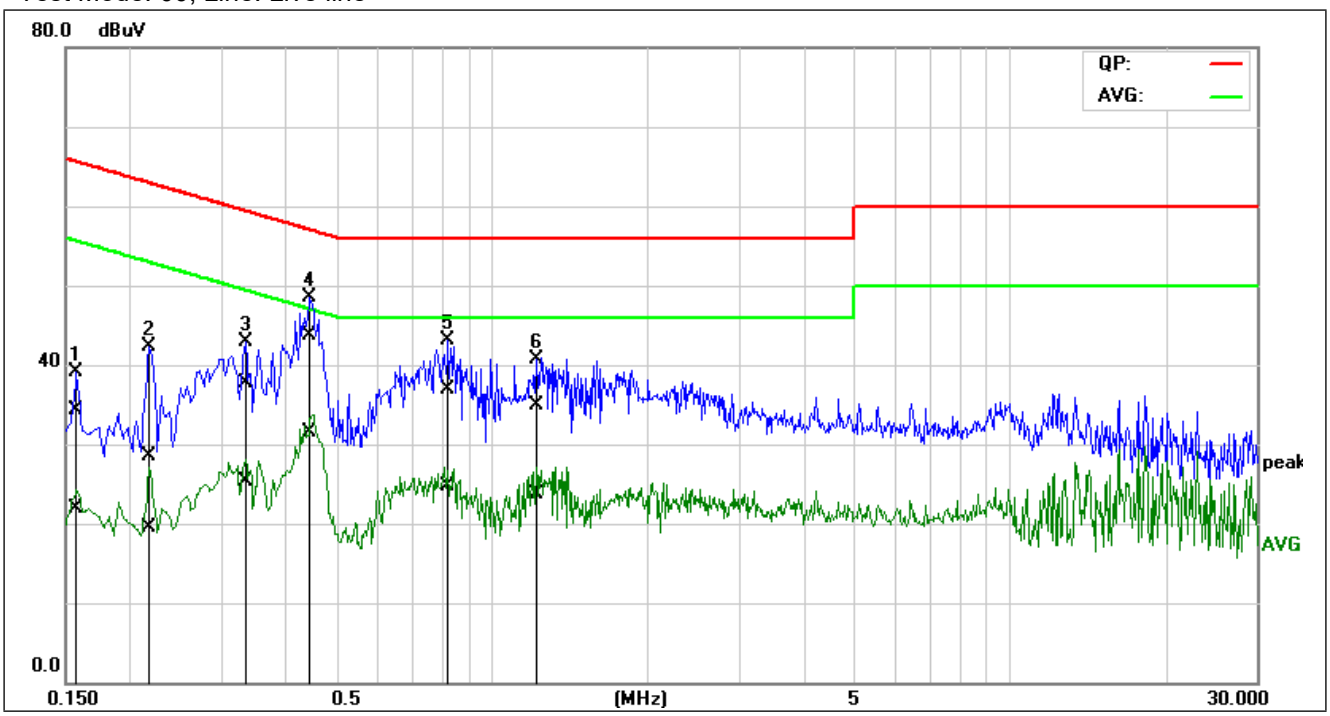
The red line show in graphic is the limit in standard used in this section.

Measured Level = Read level + Cable Loss + LISN Factor

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Test Mode: 00; Line: Live line

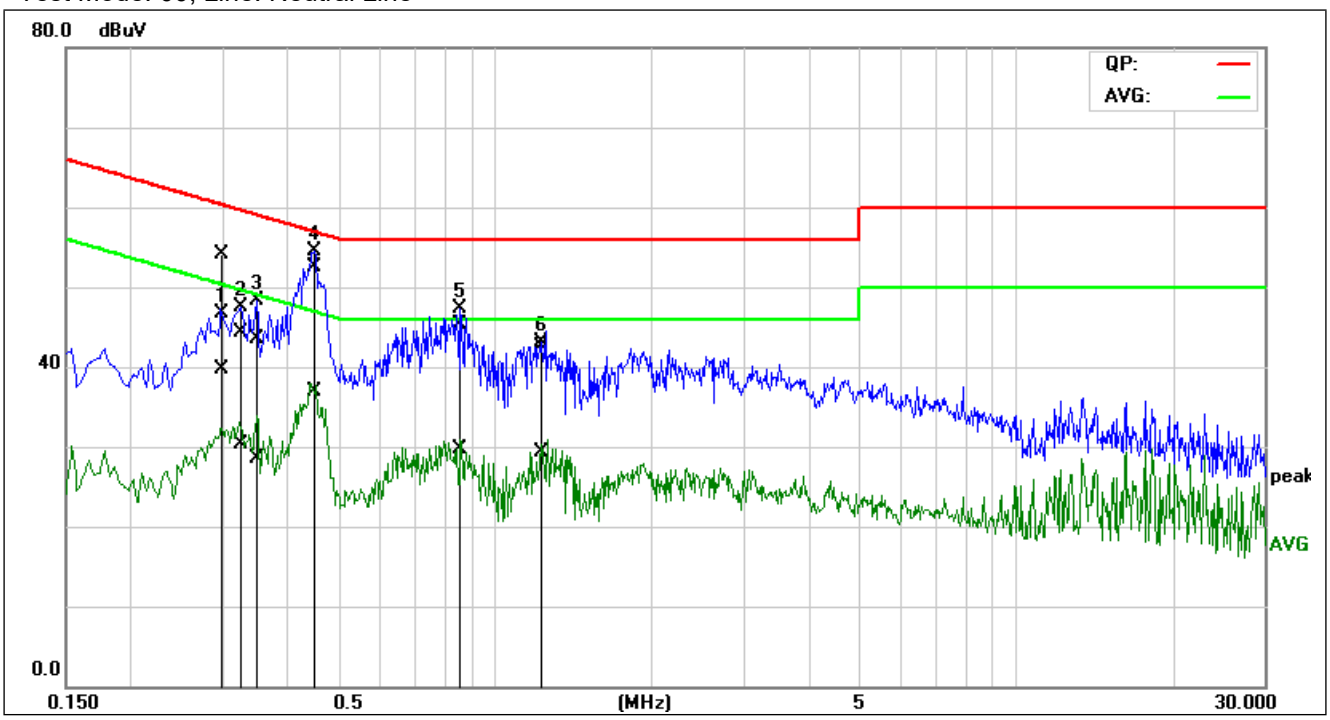



| No. | Frequency | QuasiPeak reading | Average reading | Correction factor | QuasiPeak result | Average result | QuasiPeak limit | Average limit | QuasiPeak margin | Average margin | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | (MHz) | (dBuV) | (dBuV) | (dB) | (dBuV) | (dBuV) | (dBuV) | (dBuV) | (dB) | (dB) | |
| 1 | 0.1582 | 14.79 | 2.40 | 19.46 | 34.25 | 21.86 | 65.56 | 55.56 | -31.31 | -33.70 | Pass |
| 2 | 0.2167 | 9.06 | 0.14 | 19.46 | 28.52 | 19.60 | 62.94 | 52.94 | -34.42 | -33.34 | Pass |
| 3 | 0.3326 | 18.16 | 5.85 | 19.47 | 37.63 | 25.32 | 59.39 | 49.39 | -21.76 | -24.07 | Pass |
| 4* | 0.4455 | 24.26 | 12.10 | 19.47 | 43.73 | 31.57 | 56.96 | 46.96 | -13.23 | -15.39 | Pass |
| 5 | 0.8248 | 17.48 | 5.32 | 19.48 | 36.96 | 24.80 | 56.00 | 46.00 | -19.04 | -21.20 | Pass |
| 6 | 1.2229 | 15.47 | 4.27 | 19.48 | 34.95 | 23.75 | 56.00 | 46.00 | -21.05 | -22.25 | Pass |

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Test Mode: 00; Line: Neutral Line




| No. | Frequency | QuasiPeak reading | Average reading | Correction factor | QuasiPeak result | Average result | QuasiPeak limit | Average limit | QuasiPeak margin | Average margin | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | (MHz) | (dBuV) | (dBuV) | (dB) | (dBuV) | (dBuV) | (dBuV) | (dBuV) | (dB) | (dB) | |
| 1 | 0.3013 | 34.53 | 20.17 | 19.50 | 54.03 | 39.67 | 60.21 | 50.21 | -6.18 | -10.54 | Pass |
| 2 | 0.3233 | 24.70 | 10.72 | 19.51 | 44.21 | 30.23 | 59.62 | 49.62 | -15.41 | -19.39 | Pass |
| 3 | 0.3509 | 23.98 | 9.01 | 19.51 | 43.49 | 28.52 | 58.94 | 48.94 | -15.45 | -20.42 | Pass |
| 4* | 0.4521 | 32.96 | 17.36 | 19.55 | 52.51 | 36.91 | 56.84 | 46.84 | -4.33 | -9.93 | Pass |
| 5 | 0.8546 | 25.72 | 10.08 | 19.59 | 45.31 | 29.67 | 56.00 | 46.00 | -10.69 | -16.33 | Pass |
| 6 | 1.2240 | 22.82 | 9.76 | 19.61 | 42.43 | 29.37 | 56.00 | 46.00 | -13.57 | -16.63 | Pass |

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## 6.2 Radiated Emission(30MHz-1GHz)

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 55011:2016+A11:2020

Limit:

| | |
|---|---|
| 30MHz-80.872MHz | 40 dB(μV/m) quasi-peak, 35 dB(μV/m) average |
| 80.872MHz-81.848MHz | 60 dB(μV/m) quasi-peak, 55 dB(μV/m) average |
| 81.848MHz-134.786MHz | 40 dB(μV/m) quasi-peak, 35 dB(μV/m) average |
| 134.786MHz-136.414MHz | 60 dB(μV/m) quasi-peak, 55 dB(μV/m) average |
| 136.414MHz-230MHz | 40 dB(μV/m) quasi-peak, 35 dB(μV/m) average |
| 230MHz-1GHz | 47 dB(μV/m) quasi-peak, 42 dB(μV/m) average |
| Detector: | Peak for pre-scan (120kHz resolution bandwidth) 30M to 1000MHz |

### 6.2.1 E.U.T. Operation

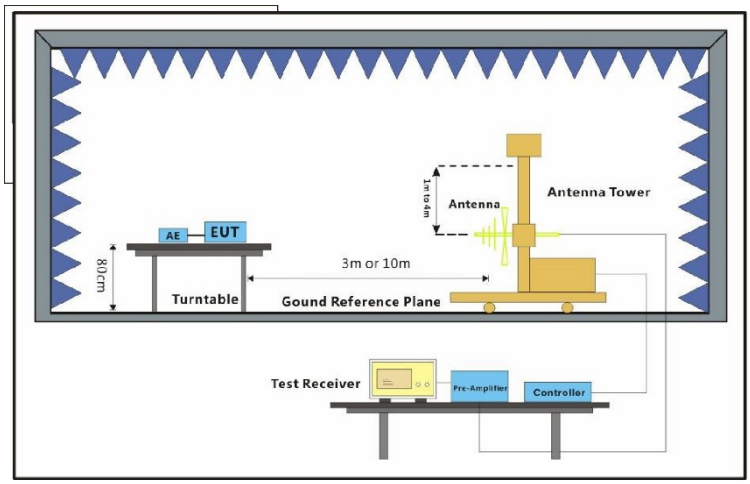
Operating Environment:

Temperature: 23.4 °C Humidity: 42.5 % RH Atmospheric Pressure: 1010 mbar

### 6.2.2 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 6.2.3 Test Setup Diagram




### 6.2.4 Measurement Procedure and Data

Frequency range: 30MHz-1GHz

An initial pre-scan was performed in the chamber using the spectrum analyser in peak detection mode. Quasi-peak measurements were conducted based on the peak sweep graph. The EUT was measured by BiConiLog antenna with 2 orthogonal polarities.

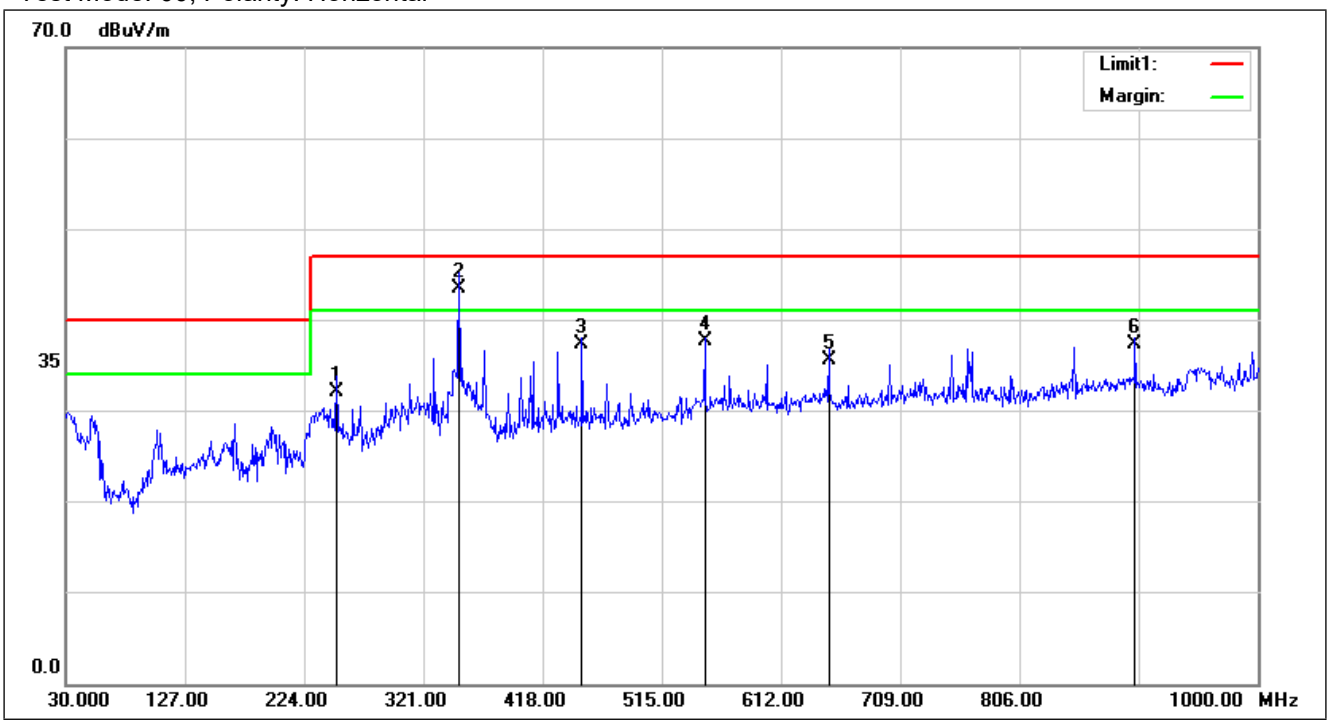
The red line show in graphic is the limit in standard used in this section.

Level=Read Level + Antenna Factor + Cable Loss - Preamp Factor

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Test Mode: 00; Polarity: Horizontal

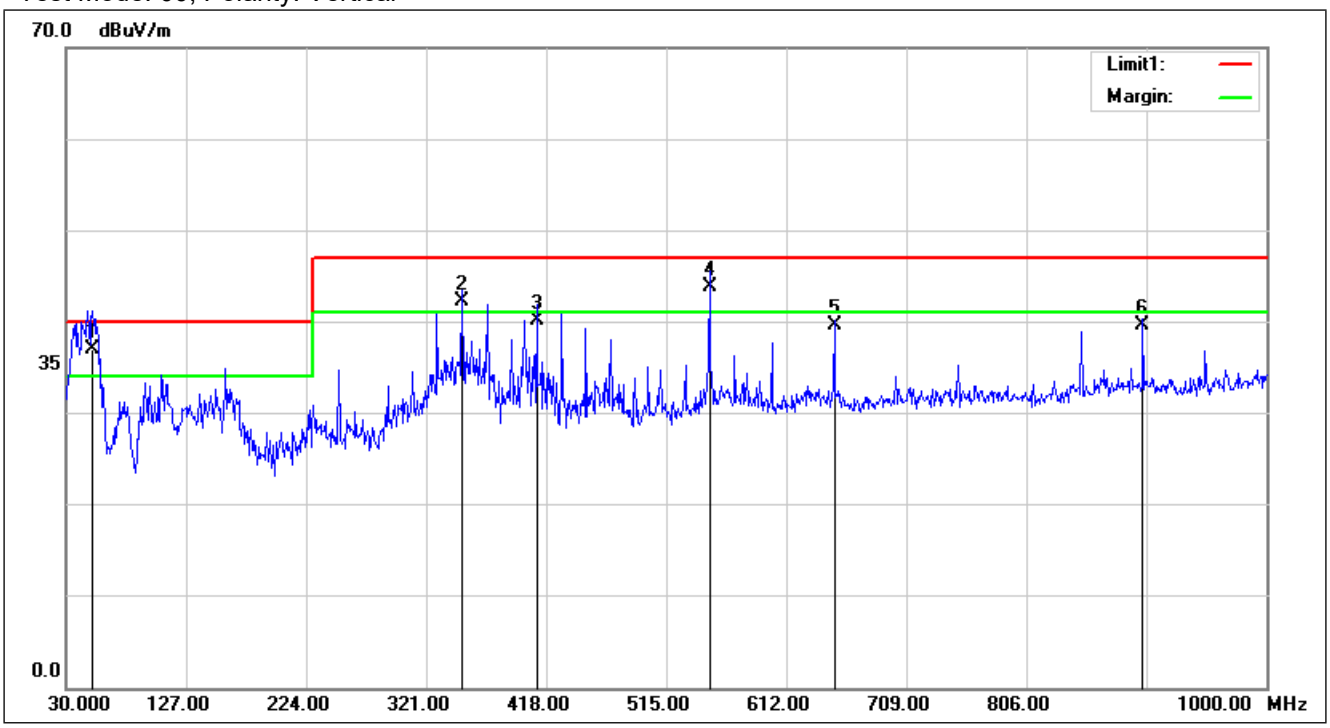



| No. | Frequency | Reading | Correct | Result | Limit | Margin | Height | Degree | Remark |
|---|---|---|---|---|---|---|---|---|---|
| | (MHz) | (dBuV) | Factor(dB/m) | (dBuV/m) | (dBuV/m) | (dB) | (cm) | (deg.) | |
| 1 | 250.1900 | 12.34 | 19.78 | 32.12 | 47.00 | -14.88 | 100 | 354 | QP |
| 2 | 350.1000 | 21.71 | 21.84 | 43.55 | 47.00 | -3.45 | 100 | 208 | QP |
| 3 | 450.0100 | 13.11 | 24.25 | 37.36 | 47.00 | -9.64 | 200 | 13 | QP |
| 4 | 549.9200 | 10.41 | 27.22 | 37.63 | 47.00 | -9.37 | 100 | 151 | QP |
| 5 | 650.8000 | 7.90 | 27.65 | 35.55 | 47.00 | -11.45 | 100 | 0 | QP |
| 6 | 900.0900 | 34.88 | 2.45 | 37.33 | 47.00 | -9.67 | 200 | 0 | QP |

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Test Mode: 00; Polarity: Vertical




| No. | Frequency | Reading | Correct | Result | Limit | Margin | Height | Degree | Remark |
|---|---|---|---|---|---|---|---|---|---|
| | (MHz) | (dBuV) | Factor(dB/m) | (dBuV/m) | (dBuV/m) | (dB) | (cm) | (deg.) | |
| 1 | 51.3400 | 19.66 | 17.30 | 36.96 | 40.00 | -3.04 | 99 | 182 | QP |
| 2 | 350.1000 | 20.38 | 21.84 | 42.22 | 47.00 | -4.78 | 201 | 246 | QP |
| 3 | 410.2400 | 16.43 | 23.79 | 40.22 | 47.00 | -6.78 | 99 | 362 | QP |
| 4 | 549.9200 | 16.61 | 27.22 | 43.83 | 47.00 | -3.17 | 201 | 343 | QP |
| 5 | 650.8000 | 11.98 | 27.65 | 39.63 | 47.00 | -7.37 | 99 | 362 | QP |
| 6 | 900.0900 | 37.18 | 2.45 | 39.63 | 47.00 | -7.37 | 99 | 362 | QP |

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## 6.3 Radiated Emissions (Magnetic field Strength)(150kHz-30MHz)

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 55011:2016+A11:2020

Limit:

0.15MHz-30MHz 39 dB(μA/m)-3 dB(μA/m) quasi-peak,

Detector: Peak for pre-scan (120kHz resolution bandwidth) 0.15M to 30MHz

### 6.3.1 E.U.T. Operation

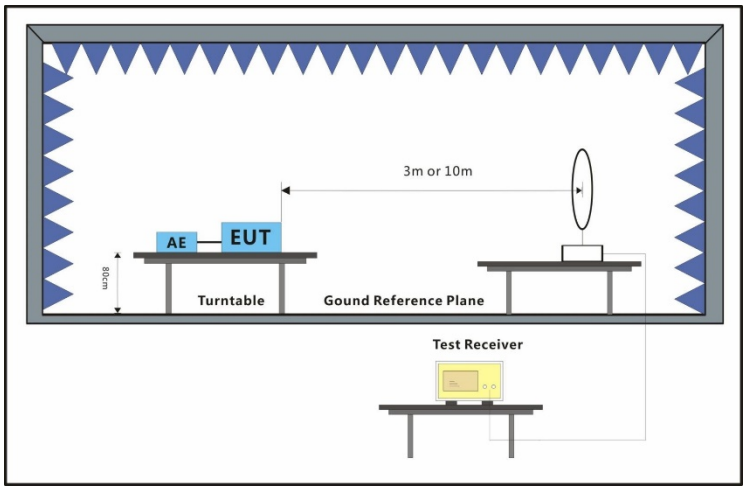
Operating Environment:

Temperature: 23.4 °C Humidity: 42.5 % RH Atmospheric Pressure: 1010 mbar

### 6.3.2 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 6.3.3 Test Setup Diagram




### 6.3.4 Measurement Procedure and Data

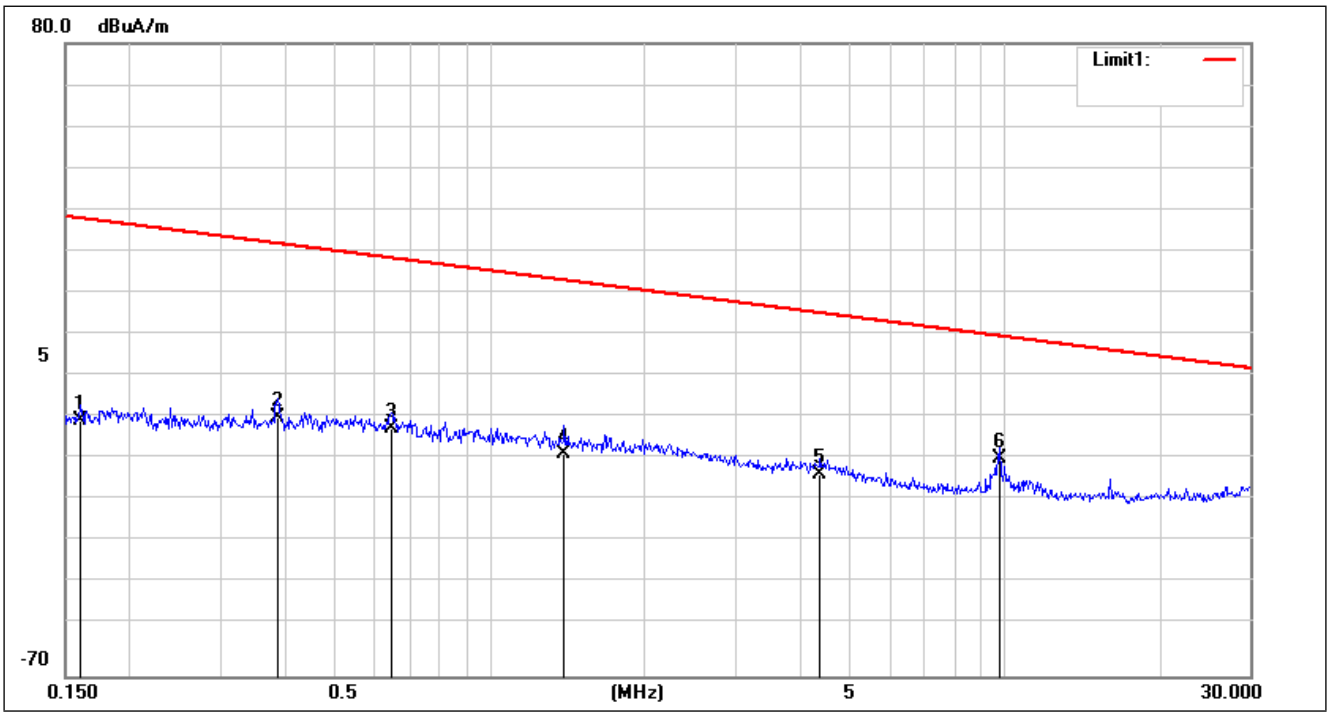
Frequency range: 150kHz-30MHz

The red line show in graphic is the limit in standard used in this section.

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Test Mode: 00; Polarity: Horizontal

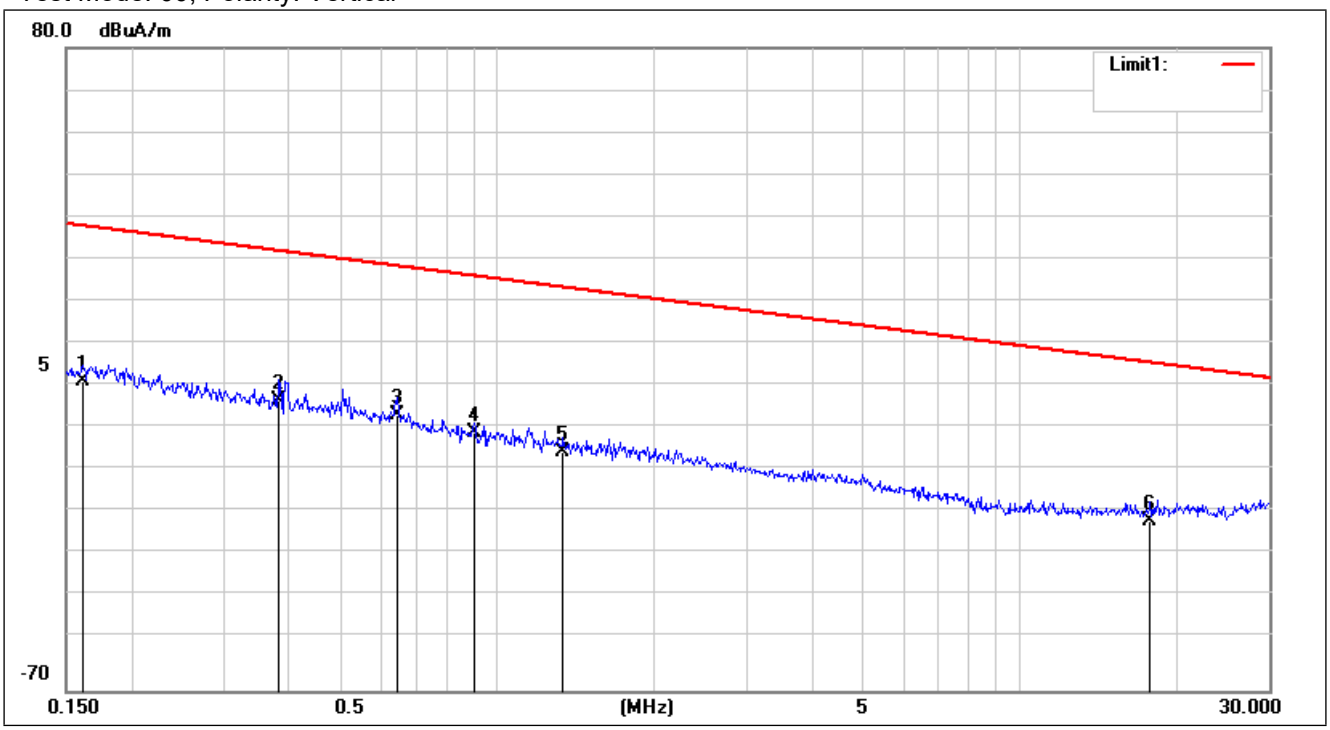



| No. | Frequency | Reading | Correct | Result | Limit | Margin | Height | Degree | Remark |
|---|---|---|---|---|---|---|---|---|---|
| | (MHz) | (dBuA/m) | Factor(dB) | (dBuA/m) | (dBuA/m) | (dB) | (cm) | (deg.) | |
| 1 | 0.1607 | 27.05 | -36.41 | -9.36 | 38.53 | -47.89 | 100 | 204 | QP |
| 2 | 0.3870 | 27.72 | -36.27 | -8.55 | 32.56 | -41.11 | 100 | 34 | QP |
| 3 | 0.6440 | 25.22 | -36.44 | -11.22 | 29.10 | -40.32 | 100 | 196 | QP |
| 4 | 1.3955 | 18.79 | -36.01 | -17.22 | 23.85 | -41.07 | 100 | 208 | QP |
| 5 | 4.3837 | 13.63 | -35.88 | -22.25 | 16.07 | -38.32 | 100 | 301 | QP |
| 6 | 9.8085 | 17.71 | -36.03 | -18.32 | 10.60 | -28.92 | 100 | 360 | QP |

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Test Mode: 00; Polarity: Vertical




| No. | Frequency | Reading | Correct | Result | Limit | Margin | Height | Degree | Remark |
|---|---|---|---|---|---|---|---|---|---|
| | (MHz) | (dBuA/m) | Factor(dB) | (dBuA/m) | (dBuA/m) | (dB) | (cm) | (deg.) | |
| 1 | 0.1615 | 38.73 | -36.41 | 2.32 | 38.50 | -36.18 | 100 | 62 | QP |
| 2 | 0.3830 | 33.95 | -36.27 | -2.32 | 32.63 | -34.95 | 100 | 360 | QP |
| 3 | 0.6440 | 30.92 | -36.44 | -5.52 | 29.10 | -34.62 | 100 | 293 | QP |
| 4 | 0.9040 | 26.70 | -36.33 | -9.63 | 26.80 | -36.43 | 100 | 46 | QP |
| 5 | 1.3380 | 21.60 | -36.03 | -14.43 | 24.13 | -38.56 | 100 | 352 | QP |
| 6 | 17.7547 | 5.55 | -35.88 | -30.33 | 6.56 | -36.89 | 100 | 296 | QP |

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## 6.4 Harmonic Current Emission

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN IEC 61000-3-2:2019+A1:2021

### 6.4.1 E.U.T. Operation

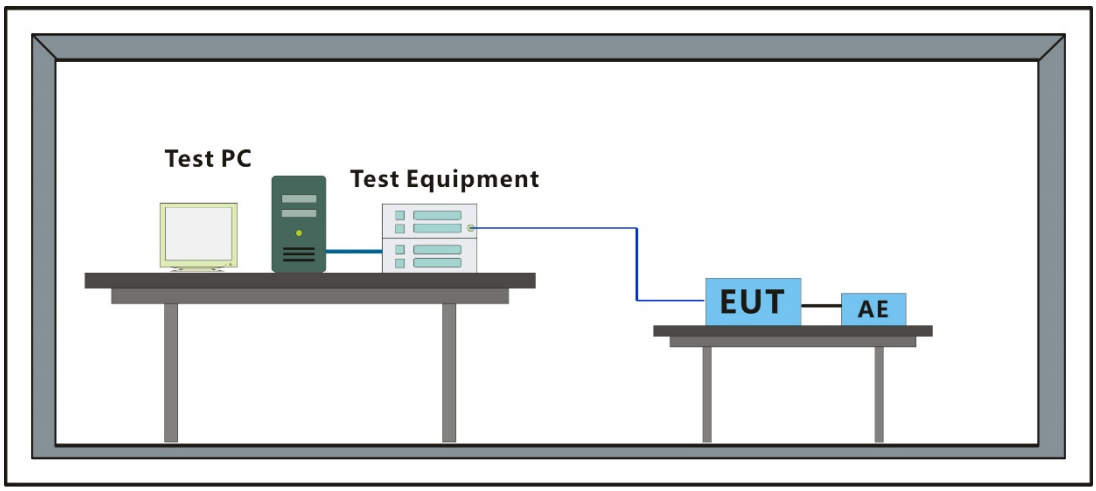
Operating Environment:

Temperature: 24 °C Humidity: 48 % RH Atmospheric Pressure: 1010 mbar

### 6.4.2 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 6.4.3 Test Setup Diagram




### 6.4.4 Measurement Procedure and Data

Frequency Range: 100Hz to 2kHz

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Test Mode: 00

Highest parameter values during test:

| | | | |
|---|---|---|---|
| V_RMS (Volts): | 230.21 | Frequency(Hz): | 50.00 |
| I_Peak (Amps): | 0.144 | I_RMS (Amps): | 0.021 |
| I_Fund (Amps): | 0.009 | Crest Factor: | 7.034 |
| Power (Watts): | 2.0 | Power Factor: | 0.435 |

| Harm# | Harms(avg) | 100%Limit | %of Limit | Harms(max) | 150%Limit | %of Limit | Status |
|---|---|---|---|---|---|---|---|
| 2 | 0.001 | 1.080 | N/A | 0.002 | 1.620 | N/A | Pass |
| 3 | 0.007 | 2.300 | 0.3 | 0.008 | 3.450 | 0.2 | Pass |
| 4 | 0.001 | 0.430 | N/A | 0.001 | 0.645 | N/A | Pass |
| 5 | 0.006 | 1.140 | 0.5 | 0.007 | 1.710 | 0.4 | Pass |
| 6 | 0.001 | 0.300 | N/A | 0.001 | 0.450 | N/A | Pass |
| 7 | 0.006 | 0.770 | 0.8 | 0.007 | 1.155 | 0.6 | Pass |
| 8 | 0.001 | 0.230 | N/A | 0.001 | 0.345 | N/A | Pass |
| 9 | 0.006 | 0.400 | 1.5 | 0.006 | 0.600 | 1.0 | Pass |
| 10 | 0.001 | 0.184 | N/A | 0.001 | 0.276 | N/A | Pass |
| 11 | 0.006 | 0.330 | 1.7 | 0.006 | 0.495 | 1.2 | Pass |
| 12 | 0.000 | 0.153 | N/A | 0.001 | 0.230 | N/A | Pass |
| 13 | 0.005 | 0.210 | 2.5 | 0.005 | 0.315 | 1.7 | Pass |
| 14 | 0.000 | 0.131 | N/A | 0.001 | 0.197 | N/A | Pass |
| 15 | 0.005 | 0.150 | 3.4 | 0.005 | 0.225 | 2.3 | Pass |
| 16 | 0.000 | 0.115 | N/A | 0.001 | 0.173 | N/A | Pass |
| 17 | 0.005 | 0.132 | N/A | 0.005 | 0.198 | N/A | Pass |
| 18 | 0.000 | 0.102 | N/A | 0.001 | 0.153 | N/A | Pass |
| 19 | 0.004 | 0.118 | N/A | 0.004 | 0.178 | N/A | Pass |
| 20 | 0.000 | 0.092 | N/A | 0.000 | 0.138 | N/A | Pass |
| 21 | 0.004 | 0.107 | N/A | 0.004 | 0.161 | N/A | Pass |
| 22 | 0.000 | 0.084 | N/A | 0.000 | 0.125 | N/A | Pass |
| 23 | 0.003 | 0.098 | N/A | 0.004 | 0.147 | N/A | Pass |
| 24 | 0.000 | 0.077 | N/A | 0.000 | 0.115 | N/A | Pass |
| 25 | 0.003 | 0.090 | N/A | 0.003 | 0.135 | N/A | Pass |
| 26 | 0.000 | 0.071 | N/A | 0.000 | 0.107 | N/A | Pass |
| 27 | 0.003 | 0.083 | N/A | 0.003 | 0.125 | N/A | Pass |
| 28 | 0.000 | 0.066 | N/A | 0.000 | 0.099 | N/A | Pass |
| 29 | 0.002 | 0.078 | N/A | 0.002 | 0.116 | N/A | Pass |
| 30 | 0.000 | 0.061 | N/A | 0.000 | 0.092 | N/A | Pass |
| 31 | 0.002 | 0.073 | N/A | 0.002 | 0.109 | N/A | Pass |
| 32 | 0.000 | 0.058 | N/A | 0.000 | 0.086 | N/A | Pass |
| 33 | 0.002 | 0.068 | N/A | 0.002 | 0.102 | N/A | Pass |
| 34 | 0.000 | 0.054 | N/A | 0.000 | 0.081 | N/A | Pass |
| 35 | 0.001 | 0.064 | N/A | 0.001 | 0.096 | N/A | Pass |
| 36 | 0.000 | 0.051 | N/A | 0.000 | 0.077 | N/A | Pass |
| 37 | 0.001 | 0.061 | N/A | 0.001 | 0.091 | N/A | Pass |
| 38 | 0.000 | 0.048 | N/A | 0.000 | 0.073 | N/A | Pass |
| 39 | 0.001 | 0.058 | N/A | 0.001 | 0.087 | N/A | Pass |
| 40 | 0.000 | 0.046 | N/A | 0.000 | 0.069 | N/A | Pass |

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| Harm# | Harmonics V-rms | Limit V-rms | % of Limit | Status |
|---|---|---|---|---|
| 2 | 0.156 | 0.460 | 34.00 | OK |
| 3 | 0.447 | 2.072 | 21.56 | OK |
| 4 | 0.056 | 0.460 | 12.18 | OK |
| 5 | 0.045 | 0.921 | 4.87 | OK |
| 6 | 0.041 | 0.460 | 9.01 | OK |
| 7 | 0.041 | 0.690 | 5.97 | OK |
| 8 | 0.031 | 0.460 | 6.74 | OK |
| 9 | 0.030 | 0.460 | 6.57 | OK |
| 10 | 0.025 | 0.460 | 5.39 | OK |
| 11 | 0.016 | 0.230 | 7.02 | OK |
| 12 | 0.018 | 0.230 | 7.69 | OK |
| 13 | 0.016 | 0.230 | 6.99 | OK |
| 14 | 0.009 | 0.230 | 4.12 | OK |
| 15 | 0.010 | 0.230 | 4.15 | OK |
| 16 | 0.015 | 0.230 | 6.44 | OK |
| 17 | 0.013 | 0.230 | 5.71 | OK |
| 18 | 0.016 | 0.230 | 6.75 | OK |
| 19 | 0.010 | 0.230 | 4.54 | OK |
| 20 | 0.009 | 0.230 | 3.85 | OK |
| 21 | 0.012 | 0.230 | 5.31 | OK |
| 22 | 0.007 | 0.230 | 2.93 | OK |
| 23 | 0.010 | 0.230 | 4.36 | OK |
| 24 | 0.007 | 0.230 | 2.89 | OK |
| 25 | 0.011 | 0.230 | 4.86 | OK |
| 26 | 0.010 | 0.230 | 4.55 | OK |
| 27 | 0.008 | 0.230 | 3.67 | OK |
| 28 | 0.008 | 0.230 | 3.38 | OK |
| 29 | 0.008 | 0.230 | 3.66 | OK |
| 30 | 0.007 | 0.230 | 2.99 | OK |
| 31 | 0.006 | 0.230 | 2.69 | OK |
| 32 | 0.006 | 0.230 | 2.42 | OK |
| 33 | 0.007 | 0.230 | 2.87 | OK |
| 34 | 0.004 | 0.230 | 1.52 | OK |
| 35 | 0.004 | 0.230 | 1.89 | OK |
| 36 | 0.003 | 0.230 | 1.52 | OK |
| 37 | 0.005 | 0.230 | 2.01 | OK |
| 38 | 0.003 | 0.230 | 1.13 | OK |
| 39 | 0.004 | 0.230 | 1.52 | OK |
| 40 | 0.004 | 0.230 | 1.94 | OK |

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## 6.5 Voltage Fluctuations and Flicker

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 61000-3-3:2013+A1:2019

### 6.5.1 E.U.T. Operation

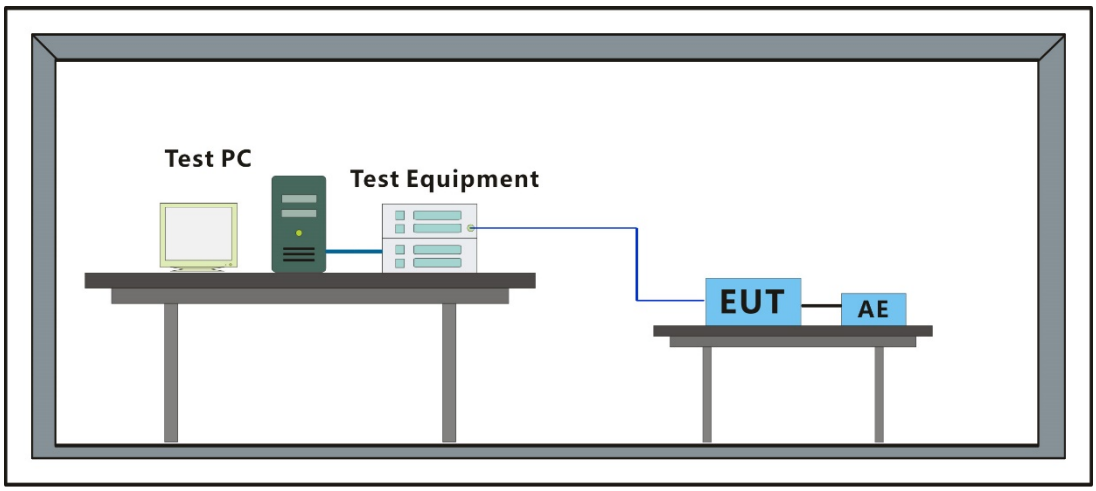
Operating Environment:

Temperature: 24 °C Humidity: 48 % RH Atmospheric Pressure: 1010 mbar

### 6.5.2 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 6.5.3 Test Setup Diagram




### 6.5.4 Measurement Procedure and Data

Test Mode: 00

Parameter values recorded during the test:

| | | | | |
|---|---|---|---|---|
| Vrms at the end of test (Volt): | 230.27 | | | |
| T-max (mS): | 0 | Test limit (mS): | 500.0 | Pass |
| Highest dc (%): | 0.00 | Test limit (%): | 3.30 | Pass |
| Highest dmax (%): | 0.00 | Test limit (%): | 4.00 | Pass |
| Highest Pst (10 min. period): | 0.064 | Test limit: | 1.000 | Pass |
| Highest Plt (2 hr. period): | 0.028 | Test limit: | 0.650 | Pass |

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# 7 Immunity Test Results

**Performance Criteria Description in EN IEC 61326-1:2021**

| | |
|---|---|
| **Criterion A** | The equipment shall continue to operate as intended during and after the test. No DEGRADATION OF PERFORMANCE or LOSS OF FUNCTION is allowed below a PERFORMANCE LEVEL specified in the user documentation, when the equipment is used as intended. In the case of applying immunity tests with continuous electromagnetic phenomena, the PERFORMANCE LEVEL may be replaced by a permissible LOSS OF PERFORMANCE which shall recover, without user intervention. A permissible LOSS OF PERFORMANCE is allowed within the PERFORMANCE LEVEL only when this information is clearly provided to the end user via documentation, such as the product user manual. No change in the operating state is allowed nor is loss of data. |
| **Criterion B** | The equipment shall continue to operate as intended after the test. No DEGRADATION OF PERFORMANCE or LOSS OF FUNCTION is allowed below a PERFORMANCE LEVEL specified in the user documentation, when the equipment is used as intended. During the test, the equipment PERFORMANCE LEVEL may be replaced by a permissible LOSS OF PERFORMANCE if such LOSS OF PERFORMANCE is detailed in the EMC test plan. A permissible LOSS OF PERFORMANCE is allowed within the PERFORMANCE LEVEL only when this information is clearly provided to the end user via documentation, such as the product user manual. An unintended change of the operating state is allowed if self-recoverable. No loss of stored data is allowed. |
| **Criterion C** | LOSS OF FUNCTION is allowed, provided the function is self-recoverable or can be restored by the operation of the controls. Recovery procedure shall be included in the user documentation. No permanent damage to the equipment is allowed. |

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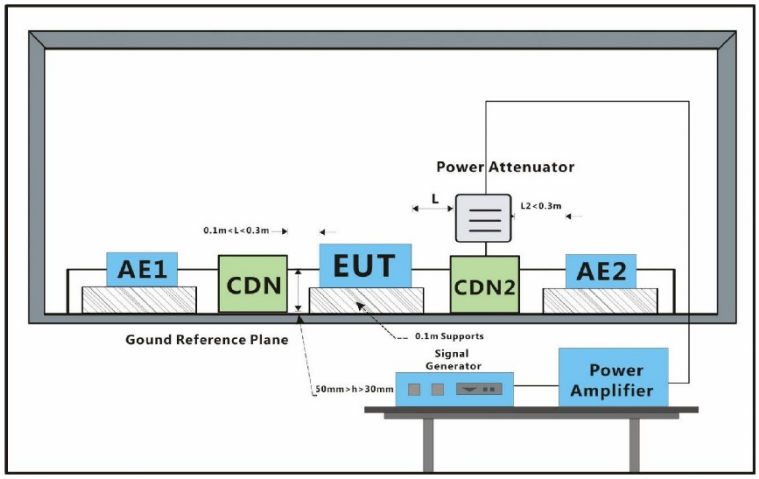
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## 7.1 Conducted Immunity at AC Mains Power Port (150kHz-80MHz)

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 61000-4-6:2014

### 7.1.1 Test Setup Diagram




### 7.1.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

### 7.1.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.1.4 Test Condition and Results:

Performance Criterion: A

Frequency Range: 0.15MHz to 80MHz

Modulation: 80%, 1kHz Amplitude Modulation

Step Size: 1%

| Cable port | Level (Vrms) | CDN/Clamp | Dwell time | Result / Observations |
|---|---|---|---|---|
| AC power port | 3(0.15MHz-80MHz) | CDN | 3s | A |
| A: No degradation in the performance of the EUT was observed | | | | |

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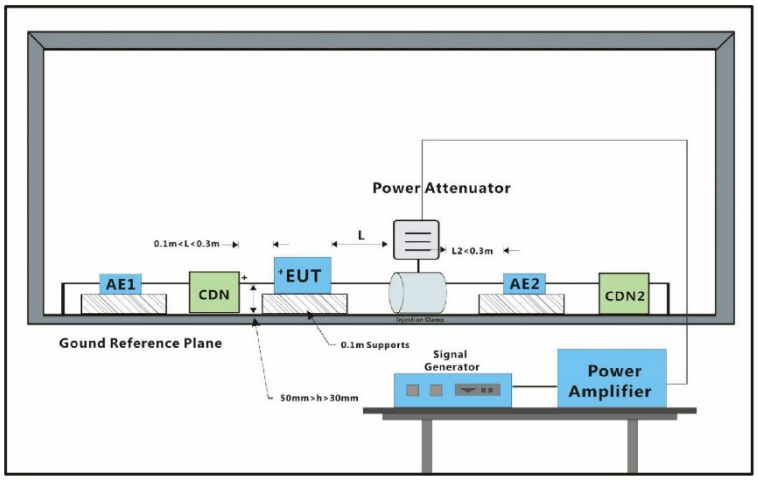
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## 7.2 Conducted Immunity at Signal Port (150kHz-80MHz)

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 61000-4-6:2014

### 7.2.1 Test Setup Diagram




### 7.2.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

### 7.2.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.2.4 Test Condition and Results:

Performance Criterion: A

Frequency Range: 0.15MHz to 80MHz

Modulation: 80%, 1kHz Amplitude Modulation

Step Size: 1%

| Cable port | Level (Vrms) | CDN/Clamp | Dwell time | Result / Observations |
|---|---|---|---|---|
| signal line | 3(0.15MHz-80MHz) | CDN | 3s | A |
| A: No degradation in the performance of the EUT was observed | | | | |

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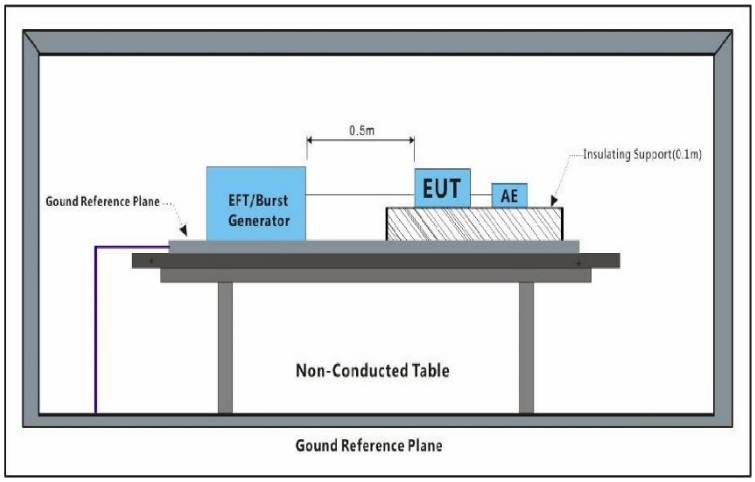
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## 7.3 Electrical Fast Transients Burst at AC Mains Power Port

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 61000-4-4:2012

### 7.3.1 Test Setup Diagram




### 7.3.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

### 7.3.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.3.4 Test Condition and Results:

Performance Criterion:B

Repetition Frequency:5kHz

Burst Period:300ms

Test Duration:2 minute per level & polarity

Test Level:1.0kV

Polarity:Positive & Negative

| Test Line | Level (kV) | Polarity | CDN/Clamp | Result / Observations |
|---|---|---|---|---|
| AC power port | 1 | + | CDN | A |
| AC power port | 1 | - | CDN | A |
| A: No degradation in the performance of the EUT was observed | | | | |

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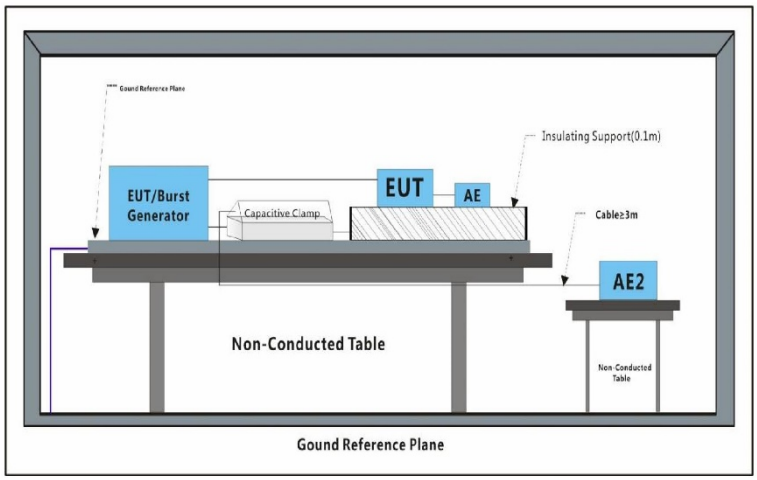
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## 7.4 Electrical Fast Transients&Burst at Signal Port

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 61000-4-4:2012

### 7.4.1 Test Setup Diagram




### 7.4.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

### 7.4.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.4.4 Test Condition and Results:

Performance Criterion: B

Repetition Frequency: 5kHz

Burst Period: 300ms

Test Duration: 2 minute per level & polarity

Polarity: Positive & Negative

| Test Line | Level (kV) | Polarity | CDN/Clamp | Result / Observations |
|---|---|---|---|---|
| Signal port | 1 | + | Clamp | A |
| Signal port | 1 | - | Clamp | A |
| A: No degradation in the performance of the EUT was observed | | | | |

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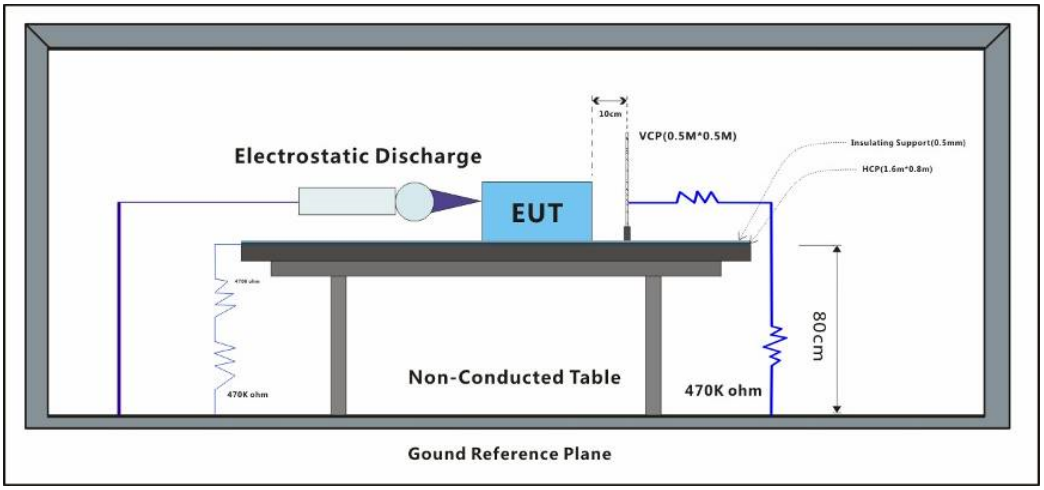
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## 7.5 Electrostatic Discharge

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 61000-4-2:2009

### 7.5.1 Test Setup Diagram




### 7.5.2 E.U.T. Operation

Operating Environment:

Temperature: 23.3 °C Humidity: 48.0 % RH Atmospheric Pressure: 1010 mbar

### 7.5.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.5.4 Test Condition and Results:

Performance Criterion:B

Discharge Impedance:330 Ω / 150 pF

Discharge Voltage:Air Discharge:8 kV,Contact Discharge:4 kV, VCP/HCP:4 kV

Polarity:Positive & Negative

Number of Discharge:Minimum 10 times at each test point

Discharge Mode:Single Discharge

Discharge Period:1 second minimum

Test Point 1: All insulated enclosure & seams.

Test Point 2: All accessible metal parts of the enclosure.

Test Point 3: All sides.

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| Discharge type | Level (kV) | Polarity | Test Point | Result / Observations |
|---|---|---|---|---|
| Air Discharge | 2,4,8 | + | 1 | A |
| Air Discharge | 2,4,8 | - | 1 | A |
| Contact Discharge | 4 | + | 2 | A |
| Contact Discharge | 4 | - | 2 | A |
| Horizontal Coupling | 4 | + | 3 | A |
| Horizontal Coupling | 4 | - | 3 | A |
| Vertical Coupling | 4 | + | 3 | A |
| Vertical Coupling | 4 | - | 3 | A |
| A: No degradation in the performance of the EUT was observed | | | | |

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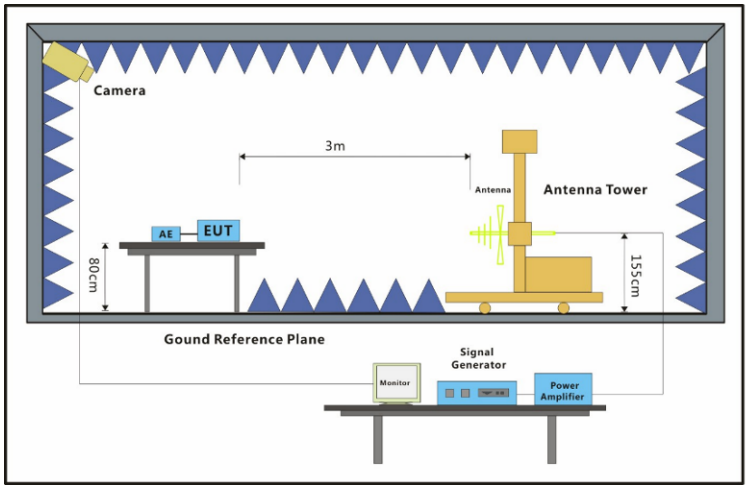
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## 7.6 Radiated Immunity(80MHz-6GHz)

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN IEC 61000-4-3: 2020

### 7.6.1 Test Setup Diagram




### 7.6.2 E.U.T. Operation

Operating Environment:

Temperature: 22.4 °C Humidity: 46..5 % RH Atmospheric Pressure: 1010 mbar

### 7.6.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.6.4 Test Condition and Results:

Performance Criterion:A

Frequency Range:80MHz to 1GHz, 1.4GHz to 6GHz.

Antenna Polarisation:Vertical and Horizontal

Modulation:1kHz,80% Amp. Mod,1% increment

Test Distance:3m

| Frequency | Level (V/m) | EUT Face | Dwell time | Result / Observations |
|---|---|---|---|---|
| 80MHz-1GHz | 3 | Front | 3s | A |
| 80MHz-1GHz | 3 | Back | 3s | A |
| 80MHz-1GHz | 3 | Left | 3s | A |
| 80MHz-1GHz | 3 | Right | 3s | A |
| 80MHz-1GHz | 3 | Top | 3s | A |
| 80MHz-1GHz | 3 | Underside | 3s | A |
| 1.4GHz-6GHz | 3 | Front | 3s | A |
| 1.4GHz-6GHz | 3 | Back | 3s | A |
| 1.4GHz-6GHz | 3 | Left | 3s | A |

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| | | | | |
|---|---|---|---|---|
| 1.4GHz-6GHz | 3 | Right | 3s | A |
| 1.4GHz-6GHz | 3 | Top | 3s | A |
| 1.4GHz-6GHz | 3 | Underside | 3s | A |
| A: No degradation in the performance of the EUT was observed | | | | |

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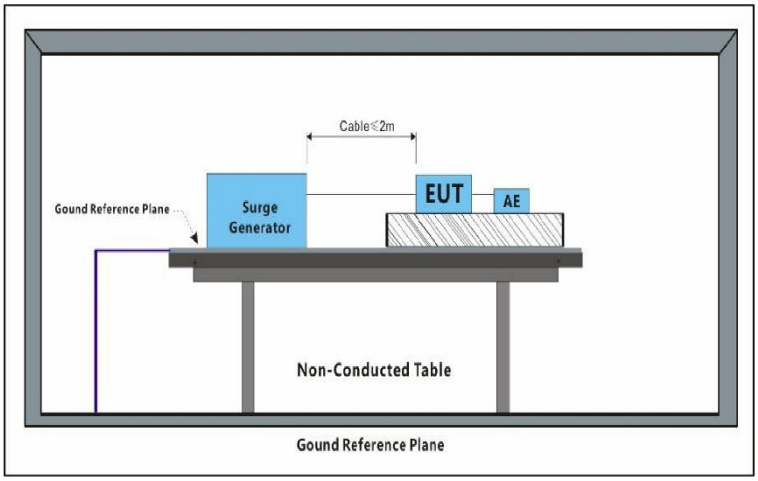
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## 7.7 Surge at AC Mains Power Port

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 61000-4-5:2014 +A1:2017

### 7.7.1 Test Setup Diagram




### 7.7.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

### 7.7.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.7.4 Test Condition and Results:

Performance Criterion:B

Interval:60s between each surge

Polarity:Positive & Negative

Generator source impedance:2Ω

CDN coupling impedance(Line-to-ground):10Ω

Trigger Mode:Internal

No. of surges:5 positive at 0°,90°,180°,270° 5 negative at 0°,90°,180°,270°.

Test Level:±0.5, 1kV Live to Neutral, ±1, 2kV Live, Neutral to Earth

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| Test Line | Level (kV) | Polarity | Phase (deg) | Result / Observations |
|---|---|---|---|---|
| L-N | 0.5 | + | 0° | A |
| L-N | 0.5 | - | 0° | A |
| L-N | 0.5 | + | 90° | A |
| L-N | 0.5 | - | 90° | A |
| L-N | 0.5 | + | 180° | A |
| L-N | 0.5 | - | 180° | A |
| L-N | 0.5 | + | 270° | A |
| L-N | 0.5 | - | 270° | A |
| A: No degradation in the performance of the EUT was observed | | | | |

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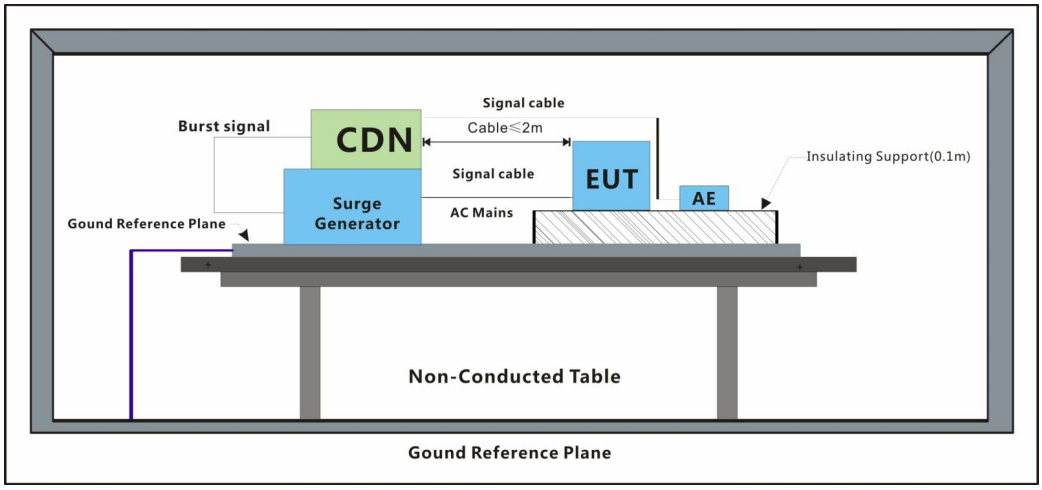
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## 7.8 Surge at Signal Port

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 61000-4-5:2014 +A1:2017

### 7.8.1 Test Setup Diagram




### 7.8.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

### 7.8.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.8.4 Test Condition and Results:

Performance Criterion: B

Interval: 60s between each surge

Polarity: Positive & Negative

Generator source impedance: 2Ω

CDN coupling impedance(Line-to-ground):10Ω

Trigger Mode: Internal

No. of surges: 5 positive 5 negative.

| Test Line | Level (kV) | Polarity | Result / Observations |
|---|---|---|---|
| Line to line (Signal) | 0.5 | + | A |
| Line to line (Signal) | 0.5 | - | A |
| Line to ground (Signal) | 1 | + | A |
| Line to ground (Signal) | 1 | - | A |
| A: No degradation in the performance of the EUT was observed | | | |

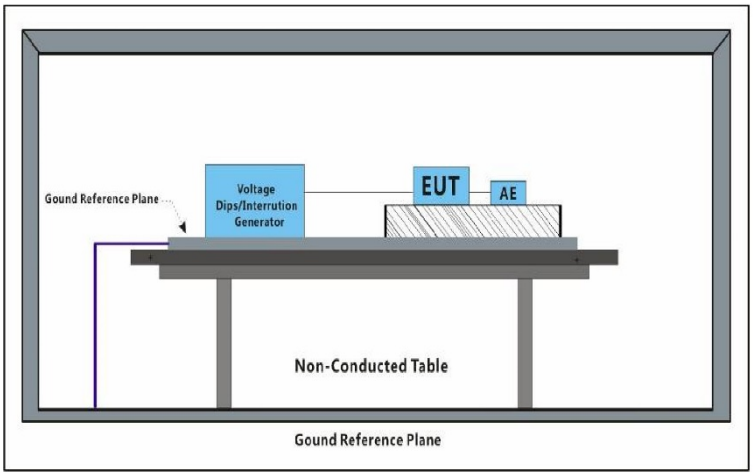
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## 7.9 Voltage Dips and Interruptions

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN IEC 61000-4-11:2020

### 7.9.1 Test Setup Diagram




### 7.9.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

### 7.9.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.9.4 Test Condition and Results:

Performance Criterion:

0% of UT (Supply Voltage) for 1 Cycle:B;

40% of UT for 10 Cycles:C;

70% of UT for 25 Cycles:C;

0% of UT for 250 Cycles:C;

No. of Dips / Interruptions:3 per Level

Time between dropout:10s

| Level % UT | Phase (deg) | Duration | No. of Dips / Interruptions | Result / Observations |
|---|---|---|---|---|
| 0 | 0° | 0.5 Cycle | 3 | A |
| 0 | 180° | 0.5 Cycle | 3 | A |
| 0 | 0° | 1 Cycle | 3 | A |
| 0 | 180° | 1 Cycle | 3 | A |
| 70 | 0° | 25 Cycles/30 Cycles | 3 | A |
| 70 | 180° | 25 Cycles/30 Cycles | 3 | A |

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| | | | | |
|---|---|---|---|---|
| 0 | 0° | 250 Cycles/300 Cycles | 3 | C |
| 0 | 180° | 250 Cycles/300 Cycles | 3 | C |
| A: No degradation in the performance of the EUT was observed<br>C: During the test EUT stop working when power supply drop,<br>After the test by manual operation can work normally. | | | | |

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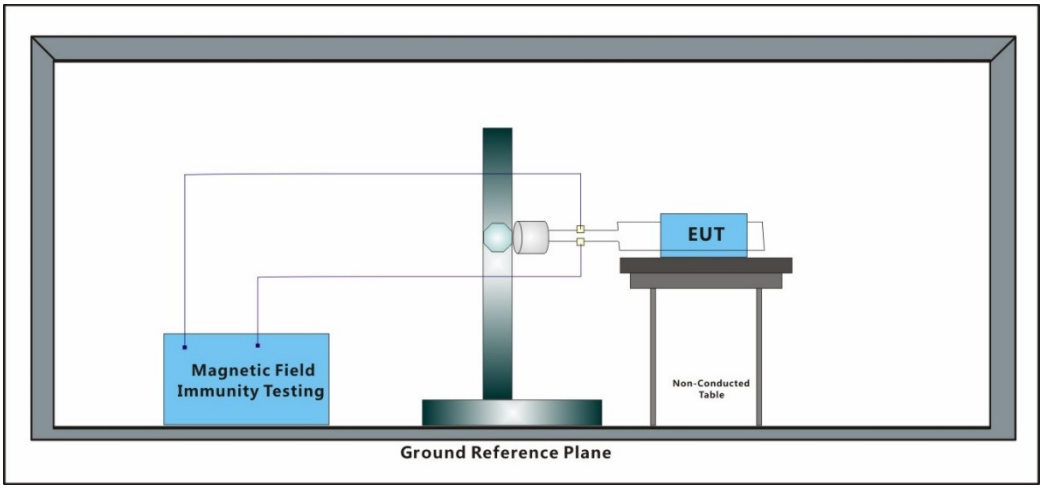
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## 7.10 Power Frequency Magnetic Field

Test Requirement: EN IEC 61326-1: 2021

Test Method: EN 61000-4-8:2010

### 7.10.1 Test Setup Diagram




### 7.10.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

### 7.10.3 Test Mode Description

| Pre-scan / Final test | Mode Code | Description |
|---|---|---|
| Final test | 00 | Keep EUT working continuously with Auxiliary equipment |

### 7.10.4 Test Condition and Results:

Performance Criterion: A

| Frequency | Level (A/m) | Axial | Magnetic Field Type | Result / Observations |
|---|---|---|---|---|
| 50Hz/60Hz | 3 | X | Continuous filed | A |
| 50Hz/60Hz | 3 | Y | Continuous filed | A |
| 50Hz/60Hz | 3 | Z | Continuous filed | A |
| A: No degradation in the performance of the EUT was observed | | | | |

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# 8 Test Setup Photo

Refer to Appendix - Test Setup Photo for KSEM2204000431AT

# 9 EUT Constructional Details (EUT Photos)

Refer to Appendix - Photographs of EUT Constructional Details for KSEM2204000431AT

- End of the Report -

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